Industrial Marketing Management 42 (2013) 125–134

Contents lists available at SciVerse ScienceDirect

Industrial Marketing Management

Spanning organizational boundaries to manage creative processes: The case of the LEGO Group

Poul Houman Andersen [a,*], Hanne Kragh [a], Christopher Lettl [b]

[a] Aarhus University, Business and Social Sciences, Dept. of Business Administration, Bartholins Alle 10, DK-8000 Aarhus C, Denmark
[b] Vienna University of Economics and Business, Dept. of Strategic Management and Innovation, Nordbergstraße 15, 1090 Vienna, Austria

ARTICLE INFO

Article history:
Received 21 June 2012
Received in revised form 24 October 2012
Accepted 24 October 2012
Available online 8 December 2012

Keywords:
Boundary spanning
Organizational creativity
Inter-organizational collaboration
Relationship management

ABSTRACT

In order to continue to be innovative in the current fast-paced and competitive environment, organizations are increasingly dependent on creative inputs developed outside their boundaries. The paper addresses the boundary spanning activities that managers undertake to a) select and mobilize creative talent, b) create shared identity, and c) combine and integrate knowledge in innovation projects involving external actors. We study boundary spanning activities in two creative projects in the LEGO group. One involves identifying and integrating deep, specialized knowledge, the other focuses on the use of external actors as a source of broad, not necessarily fully developed ideas. We find that the boundary spanning activities in these two projects differ in respect, among other things, of how the firm selects participants, formulates problems, and aligns the expectations of internal and external actors, and how knowledge is integrated across organizational boundaries. We discuss implications of our findings for managers and researchers in a business-to-business context.

1. Introduction

In step with increasing global competition, innovation capabilities are moving to the fore in a growing number of industries. As competition intensifies, the life cycle of new market offerings is decreasing, calling for firms to increase their creative capacity. Creativity can be understood as the generation of new and valuable ideas for products, processes, services, and procedures (Martins & Terblanche, 2003). Creative ideas fuel innovation, for this reason, imaginative thinking and creativity are hailed as decisive sources of competitive advantage (Florida, 2002). Due to the recombinant nature of innovation, creativity tends to require the blending of diverse knowledge bases in novel ways (Fleming, 2001). In the current business environment, knowledge is widely distributed, making it difficult for organizations to be creative without external inputs.

Involving external actors allows the organization to adopt new perspectives and apply "fresh eyes" to given problems (Jeppesen & Lakhani, 2010). It therefore can help firms to overcome organizational inertia and functional fixedness. Organizations may be able to establish networks consisting of elite circles of external actors capable of contributing significantly to and sustaining the firm's creative processes (Pisano & Verganti, 2008). From the point of view of efficiency and effectiveness and under specific conditions, shifting the locus of problem solving to external actors may be superior to traditional internal methods. These conditions include the modularity of the problems to be solved and the distribution of the problem solving knowledge (Afuah & Tucci, 2012). Many organizations set out to involve actors external to the firm in their creative activities. However, firms may have fundamentally different objectives (Terwiesch & Xu, 2008). For example, they may be searching for a range of concrete and path breaking *solutions* whose development may involve trial and error, experimentation, high levels of specialization, and high upfront investment from the external partner(s). This involves the external problem solver in revealing real and profound proprietary intellectually property. Alternatively, organizations may want to tap into a broad range of *ideas* (as opposed to fully-elaborated solutions) to gain inspiration and open up potentially attractive business opportunities. These ideas may be rather fuzzy in nature and require some experimentation and some small upfront investments by the external actors. In this case, the knowledge revealed by the external partners will be fairly superficial.

When tapping into creative ideas outside their boundaries, organizations face certain challenges, which, depending on the type of creative search (i.e. for solutions vs. ideas) will require different solutions. First, the organization must carefully and deliberately balance two potentially conflicting objectives. The external input should provide perspectives which go (far) beyond existing firm perspectives, but which at the same time, are aligned to the firm's corporate strategy. Second, the problem to be addressed by the external actors should not be too narrowly framed since this might preclude creativity, but neither should its framing be too broad since this might blur the problem statement. Third, there are important issues related to designing the

* Corresponding author. Tel.: +45 51 94 18 83.
E-mail addresses: poa@asb.dk (P.H. Andersen), hak@asb.dk (H. Kragh), christopher.lettl@wu.ac.at (C. Lettl).

0019-8501/$ – see front matter © 2012 Elsevier Inc. All rights reserved.
http://dx.doi.org/10.1016/j.indmarman.2012.11.011

incentives for participation and transfer of knowledge, and potential intellectual property rights conflicts (Franke, Keinz, & Klausberger, forthcoming). Fourth, the host organization needs to overcome the not-invented-here syndrome, prevalent in many research and development departments (Lichtenthaler & Ernst, 2006). Fifth, since the external partner may have no experience of collaboration with the organization, it may be necessary to establish a level of trust. Trust has been shown to be an important prerequisite for inter-organizational collaboration and learning (e.g. Janowicz, Panjaitan, & Krishnan, 2009; Zaheer, McEvily, & Perrone, 1998).

Boundary spanning activity is crucial to exploit opportunities and overcome the challenges inherent in the involvement of external actors in the early stages of the innovation process. Boundary spanning has been described as "behaviors intended to establish relationships and interactions with external actors that can assist [a] team in meeting its overall objectives" (Marrone, Tesluk, & Carson, 2007:1424). Boundary spanning activities may provide a closer coupling between the organization and its environment (Ancona & Caldwell, 1992; Choi, 2002). Boundary spanning activities can facilitate knowledge transfer between organizations (Hansen, 1999), promote technological development (Rosenkopf & Nerkar, 2001), and have been shown generally to improve team effectiveness and performance (Ancona, 1990; Joshi, Pandey, & Han, 2009; Marrone et al., 2007). Effective boundary spanning may help organizations to integrate external creative inputs into internal innovation projects. However, inter-organizational boundary spanning activity in the context of creativity processes and outcomes has received little research attention. The literature on organizational boundary spanning focuses primarily on alignment, that is, how to increase coordination efficiency across organizational boundaries (Dougherty & Takacs, 2004). It tends to adopt an intra-organizational perspective focusing on the interface between teams and the organizations in which they are embedded, that is, focusing on internal rather than external boundaries (Ancona & Caldwell, 1992). The inter-firm literature, on the other hand, explicitly addresses issues related to the governance of inter-organizational relationships, but rather overlooks organizational transition such as those that underlies creativity processes (Kragh & Andersen, 2009).

Against this background, the present paper is aimed at achieving a better understanding, based on a boundary spanning perspective, of how organizations deal with the challenges that emerge when the locus of problem solving and creativity shifts to external actors. We investigate the following research questions: Which boundary spanning activities are crucial for firms wanting to tap into external sources of creativity, and how does the organization manage these activities in the context of sourcing of external solutions compared to sourcing ideas? Insights gained from addressing these questions should provide a better understanding of boundary spanning activities from the perspective of inter-organizational creativity, and help firms to tailor activities to the achievement of particular objectives. The empirical setting for our study is the LEGO Group. This company was chosen for our explorative analysis because it can be considered to be at the frontier with respect to involving external actors in collaborative creative processes. We focus on two cases: a project aimed at sourcing externally developed solutions, and a project aimed at sourcing externally developed ideas.

The remainder of the paper is organized as follows. In Section 2 we develop a preliminary conceptual framework of crucial boundary spanning activities in the context of this study. Section 3 describes our research approach and methodology, and Section 4 presents the study findings. In Section 5 we discuss the implications for theory and managerial practice.

2. Organizational creativity

Although creativity has been studied intensively from a number of different scientific perspectives, there is no single authoritative definition of the phenomenon of creativity in the literature. However, it is generally agreed that creativity involves the discovery of ideas[1] that are both *original* and *useful* to the context, and that, in a business context, these ideas will be related to products, services, processes, and/or procedures (Shalley & Gilson, 2004).

There are some specific issues related to creativity in an organizational context (Woodman, Sawyer, & Griffin, 1993). Organizational conduct implies a certain degree of managerial authority, formalization, and standardization as part of the enactment of a shared understanding, stability, and trust among the key internal and external stakeholders (Perry-Smith, 2006). As a result, and because creativity is seen as being driven by individual people with a dislike for too much structure, creativity is considered as being outside of managerial control, and as happening despite managerial intentions. Some argue that creativity is best left to informal processes of self-organizing and skunk work (Augsdorfer, 2008; Burgelman, 1983); others claim that organizational conditions (structural and social arrangements) can actively stimulate and channel creative processes (Kanter, 2000; Shalley & Gilson, 2004). Our study tends towards this second view and is concerned with what managers do to coordinate creative projects that span inter-organizational boundaries. The theoretical framework developed to understand this phenomenon is rooted in the literature on boundary spanning.

3. Managing creative processes across organizational boundaries

The majority of the literature on boundary spanning and boundary spanners belongs to two streams of research that address different levels of analysis (Marrone et al., 2007). Early work on boundary spanners, mostly within the literature on marketing, adopts a predominantly individual level of analysis. Katz and Kahn (1978) refer to role theory in discussing the dual pressure and resulting role overload on individuals operating at the periphery or boundary of organizations (Goolsby, 1992; Keller & Holland, 1975; Lysonski & Johnson, 1983; Singh, 1993). Sales and marketing personnel are positioned between the organization and its environment: They represent the firm and its internal policies to the external world while playing important roles in transferring environmental information to the organization (Aldrich & Herker, 1977). They contribute to sense-making in both the organization and its environment (Geiger & Finch, 2009).

Another stream of research addresses boundary spanning at group or team level. Here, boundary spanning is conceptualized as an aggregate, team-level phenomenon, and therefore is concerned with the relationships between the team and its external environment that enable the team to meet its overall goals (Joshi et al., 2009). Research in this stream builds on increasing recognition of the effect on organizational performance of the interdependence between creative teams and other organizational units, focusing on outward directed behaviors as opposed to the traditional dominant focus on internal team dynamics (Ancona & Caldwell, 1992). Studies show that the boundary spanning activities of teams are a significant predictor of several different organizational outcomes such as knowledge sharing, innovativeness, and team effectiveness, typically with communication or knowledge flows mediating between boundary spanning behavior and performance (Joshi et al., 2009). Some authors focus on boundary spanning as a practice, and hence are concerned more with the actions than the formal role of the boundary spanner (Levina & Vaast, 2005; Orlikowski, 2002).

Boundary spanning encompasses a wide range of potential activities and the literature provides several classifications of boundary spanning activity. Work on the individual level typically considers boundary spanning activities as co-existing, but also conflicting in

[1] Ideas can range from communicative acts in the form of sketches and plans, to physical artefacts, that express the creative intent of the creator(s).

relation to the roles that boundary spanners must perform (Aldrich & Herker, 1977; Marrone et al., 2007; Tushman, 1977). The team-level literature often describes boundary spanning activities in relation to the strategies that teams use to deal with their environment in order to optimize their performance (Ancona, 1990). The classification of boundary spanning activities in this paper reflects the main challenges implied by the research question, that is, how firms motivate external actors to participate in creativity projects, and how they ensure proper integration of external inputs within the firm of external inputs. We can identify three groups of boundary spanning activities. The first concerns *selecting and mobilizing* team members from external organizations to participate in and provide input to creativity projects. It is important to mobilize external resources without recourse to managerial authority: the activity should be based on negotiation skills (Levina & Vaast, 2005). The second activity can be described as *promoting a shared identity and common perspective* among internal and external team members. It is important to build workable coalitions among people with diverse backgrounds and interests in order to avoid not-invented-here problems related to resistance to and attitudinal problems among employees faced with ideas and inputs developed outside the organization (Katz & Allen, 1982). The third activity is *combining and integrating ideas* across different fields. Although a level of diversity in perspectives may induce creativity, too much diversity leads to breakdowns in communication since people with widely differing thought structures will tend to ignore one another or misunderstand when engaging in development activities (Kristensen, 1992). Integrating knowledge across sometimes deeply specialized knowledge bases can be a major problem for boundary spanning managers (Bechky, 2003). Fig. 1 depicts the three groups of boundary spanning activities which are elaborated below.

3.1. Selecting and mobilizing external creative talent

The first step in a boundary spanning process is identifying and mobilizing competent and motivated team members in external organizations. The identification of relevant participants for creative tasks is discussed in the literature on open and user-driven innovation (von Hippel & von Krogh, 2003). For instance, Choi (2002, p. 184) refers to scouting or "garnering information from the environment" as a crucial activity. Following their identification, these external actors need to be mobilized, which is a separate, but interdependent and critical activity. Some researchers consider this to be a representational function related to the recruitment of important external actors (Ancona & Caldwell, 1992; Marrone et al., 2007). However, most contributions consider external activities as those efforts directed towards the team's environment, and do not distinguish between the efforts undertaken within or outside the organization (Ancona & Caldwell, 1992; Choi, 2002). However, given their position in the managerial hierarchy, boundary spanners usually have more influence within than outside the organization in relation to mobilizing the actors. Their influence outside the organization will depend on the level of their social capital and negotiating and networking skills, as well as their relational power, based on the centrality of their organization in the network of their organization. Other means used to attract external talent to contribute to creative activities include exclusivity, and privileged access to like-minded peers (von Hippel & von Krogh, 2003).

3.2. Creating a shared identity and common perspective

In managing teams with members from different organizations it is important to develop and maintain a sense of shared identity within the team. The inclusion in a creative team of members with different backgrounds can enhance team performance and increase levels of innovation (Ratcheva, 2009). However, integrating team members with diverse functional and occupational backgrounds is not straightforward. Differences in experience and expertise mean that members of different occupational communities may have local understandings of the tasks to be performed by the team; in order to capitalize on the benefits that can arise from differences, managers must work to integrate these different understandings (Bechky, 2003). In inter-organizational creative teams, team members may not know one another, and may not be familiar with one another's organizational backgrounds and routines, which complicates the task of integration. Poor integration among team members may result in negative reactions and not-invented-here attitudes (Katz & Allen, 1982). Also, in the early stages of creative developments, the aims may not be well defined which leaves room for alternative interpretations.

Managers' attempts to establish some common ground and a shared identity also include externally directed activities, i.e. activities directed towards the team's environment to manage its relationship with external actors, inside as well as outside the organization (Choi, 2002; Marrone et al., 2007). Creative teams depend on their surroundings for resources, support, and inputs from different specialists for example, but the creative team also influences its context. For instance, Ancona (1990) shows that team leaders employ different strategies for interacting with their environments, ranging from passive provision of information to more active promotion of activities to its surroundings. She found that teams that exploit what she describes as probing strategies, that is, where team leaders emphasized interaction between the team and its context, were evaluated by outsiders as being the highest performers. Active management of the boundary between the team and its context conveys the organization's impression of the team, which is likely to have a positive effect on the team's commitment and identity. Thus, boundary spanning can be seen as promoting the creation of a particular shared field, involving two or more organizations or communities (Levina & Vaast, 2005). Within this field, actors pursue their joint interests, for example, a particular creativity project, and at the same time differentiate themselves from outsiders to the field such as other actors within their respective organizations (Levina & Vaast, 2005).

If team members have deeply specialized knowledge bases, linking different contexts may be particularly difficult. Although a level of differences in perspectives can induce creativity, too much diversity is a barrier to communication and thought patterns in development activities (Kristensen, 1992). A shared identity and better integration of knowledge can be promoted through the use of boundary objects (Carlile, 2002; Finch & Geiger, 2010; Kellogg, Orlikowski, & Yates, 2006; Levina & Vaast, 2005). Boundary objects can be shared (span) across different problem solving contexts and "work to

Fig. 1. Boundary spanning activities in managing creativity processes across organizational boundaries.

establish a shared context that 'sits in the middle'" (Star (1989) quoted in Carlile, 2002, p. 451).

3.3. Combining and integrating knowledge

The third group of boundary spanning activities involves the transfer and integration of knowledge in creative projects. Combining knowledge across organizational boundaries can be difficult. Rosenkopf and Nerkar (2001, p. 289) discuss what they describe as "second-order competence" or "the ability of a firm to create new knowledge through recombination of knowledge across boundaries." The resulting new ideas do not necessarily reflect shared meanings between communities, rather they represent how creativity managers enact their surroundings, de-contextualizing inputs from different communities and re-contextualizing them into new products and solutions relevant to their current creative project (Kellogg et al., 2006). As already mentioned, value is created by the integration of perspectives rather than the perspectives themselves (Grant, 1996). Consensus among team members over team goals and tasks are important for motivating and shaping the behaviors of team members (Marrone et al., 2007). Boundary spanning activities aimed at combining knowledge across boundaries include forming and enforcing group norms (Choi, 2002) and establishing a common language (Teigland & Wasko, 2003). Facilitating interpersonal interactions among members of creative teams can increase knowledge integration because more and closer interaction breeds interpersonal trust, greater openness and a willingness to engage in dialogue (Jarvenpaa & Leidner, 1999).

4. Research methodology

The criterion for selecting a case for an explorative study is that it should provide the best opportunities for gathering the most relevant data (Strauss & Corbin, 1990). The LEGO Group is a company that sees creative solutions as critical for survival. We selected two of its projects to study the company's boundary spanning activities. The "packaging project" focuses on the search for an external supplier of packaging technology; it provides a scenario in which the LEGO Group is seeking deep knowledge within a clearly defined area. The "Next Kids Tech" project relates to a search for broader and more diverse knowledge inputs for the early stages of the LEGO Group's innovation processes. These two projects exemplify differences in boundary spanning activities related to the breadth and depth of knowledge needed. Also, although both projects are run by the same company they represent very different settings with respect to the stakeholders involved and the context of the creative process, and thus provide ideal frameworks for the study of creative processes spanning organizational boundaries. The LEGO Group has a well established, strong reputation as a leading and successful producer of creative toys in a highly competitive environment (Schultz & Hatch, 2003). Its consistent performance and the respect it has attracted from business practitioners suggests that research on its management practices will be relevant and interesting to a broader audience (Helm & Klode, 2011). In addition, the LEGO Group was happy to provide access to informants and information, which also allowed us to confirm critical information (Stake, 1995).

Information was gathered primarily through interviews. Mantere (2008) describes individual interviews as one of the best sources of information when trying to understand the motivations and character of the actions that take place in highly embedded social contexts such as organizations. It allows a deeper understanding of the experiences of boundary spanners, how they are interpreted and related to specific practices. Other data sources included internal documentation in the form of PowerPoint presentations and other material.

The case studies were conducted in 2009–2012. For each case, we interviewed managers with experience of managing creative processes across organizational boundaries. The interviews and interview protocol were applied to reveal how boundary spanning practitioners put ideas into practice and also regulated the conduct of others (Hall, 1997). In most cases, two of the authors conducted the interviews jointly. All the interviews were taped and then transcribed. Although the interviewees were in different positions in the company, all were engaged in and responsible for project oriented work.

4.1. Data analysis

We listened to the recordings of the interviews and read the transcriptions in order to gain a proper understanding of the meanings conveyed, in a circular exercise in which the detail provided was contrasted with emerging and more generalized theoretical thinking (Strauss & Corbin, 1990). The data were analyzed independently by each of the authors. We used Nvivo 9 (a software program for managing qualitative data) to categorize quotes and emerging insights. Following suggestions from qualitative research, we validated both processes and outcomes. Interview transcripts and final case descriptions were provided to respective respondents for comments. The authors compared and critically discussed the findings in order to sharpen our focus. We presented our draft findings at various research workshops, and received useful comments from peers. Following the recommendations of qualitative research practice, we made the analytical process as open as was possible, using citations, tabular displays of data, etc. (Miles & Huberman, 1994). The data are presented in the form of narratives that describes the two events researched. In Section 5 we contrast the cases with respect to the three types of boundary-spanning activities and use case material to elaborate them. Contrasting is a useful approach that allows identification of underlying unique as well as common patterns and dimensions, and how they are qualified by local conditions (Miles & Huberman, 1994). Our overall findings are presented in Table 1 and elaborated in the text.

5. Case analysis

5.1. The packaging project

The LEGO packaging project was a one-off project in which the LEGO Group collaborated with a group of turnkey suppliers over the development of a new global packaging system. The LEGO Group's packaging system has always been a strategic resource and a bottleneck. LEGO boxes are high quality and an integral part of the customer experience as well as being important to retailers by showcasing the LEGO products. The packaging line must be able to respond swiftly to changes, which calls for a high level of flexibility and ability to postpone production to the latest possible time in order to incorporate trends in toy sales. The LEGO Group's packaging system had grown organically, which had resulted in a system that was very complex, performed reasonably well but had limited scope. The existing packaging line was ageing and the LEGO Group was looking for suppliers willing and able to come up with creative packaging line solutions.

5.1.1. Initiating the project

The LEGO Group decided it was time to rethink its entire packaging philosophy, to prepare for the future, and to use suppliers as creative partners. The company's view was that creative inputs from outside the group were needed in order to break free of existing ways of thinking within the LEGO Group. In 2008, the LEGO concept center, which was responsible for the packaging project, hired and appointed Morten Teglgaard as manager of its "Packaging Platform Final Pack" project. His first job was to extend the ideas already developed in the LEGO Group and align the expectations, visions, and

Table 1
Comparison of the activities of creative boundary spanners in the two case projects.

		Packaging Project	Next Kids Tech
Selecting and mobilizing creative talent	Selecting participants	Individual approaches and self-selection	Broadcasting in selected talent pools
	Signaling trustworthiness and fairness	Learning possibilities from interacting with LEGO — opportunities to make a successful bid Clear communication about intellectual property rights	Signaling interesting networking opportunities Clear communication about intellectual property rights
	Developing attractive tasks	Formulating a “stretch target” — and giving individual participants support and insights	Formulating an open task and an enjoyable social/game event
Creating shared identity and common perspective	Communicating expectations	Recurrent meetings with prospect participants and carefully designed communication strategy	LEGO brand values (in the pre-workshop phase) and facilitation during the event
	Aligning participants' and LEGO's activities	Ability to showcase ideas to LEGO team and show “working prototypes” around a boundary object	Adding competition and game rules in order to encourage rapid formation of teams and standardize outputs
Combining and integrating knowledge	Generating ideas	Emerging ideas are challenged and evaluated in a context of provocative dialogue between LEGO employees and suppliers	Rapid formulation of ideas for business projects pitched in competition
	Combining creative ideas with organizational knowledge	Combining and tailoring ideas at a deep level in the LEGO manufacturing organization	Matching and testing creative ideas with LEGO employees at a superficial level to spark further internal ideation

requirements of the main internal stakeholders, towards identifying a new packaging system that integrated critical internal knowledge. One of the visible results of this process was a briefing note that described the project's background and aims for potential suppliers. The project brief stated that:

The project objectives are to develop a modular packaging platform, with maximum two line sizes. It should encompass LEAN thinking integrated, supporting production orders down to one hour and building on existing market standards for packaging and equipment and a user friendly interface that is able to cater for global differences in technical capabilities of those operating the machine.

The brief stipulated the problem areas to be addressed, but was not in the form of a tender or specification list. The intent was to find a balance between signaling openness towards suppliers and a close focus on the problem at hand. The brief was kept relatively open to allow unexpected inputs and creativity from suppliers.

The specification is made in a way where we asked to be provoked — to make them provide their outmost. If they (suppliers) are able to surprise us or change our perspective, they are in a much better position. We have specified a stretch target for them.

Within the Group, it was agreed from the start (and communicated to suppliers), that cost was not a major criterion. The project consisted of the following phases: i) search for and assessment of potential suppliers, ii) “development and delivery” in a rapid three-month creative collaboration process, and iii) co-evolution and set-up.

5.1.2. Research and assessment phases

In the search phase, Morten Teglgaard visited several international packaging machinery technology fairs to identify and make contact with prospect suppliers. He drew also on the LEGO Group's knowledge about existing suppliers' capabilities to identify prospect participants. The idea was to entice suppliers that saw the project as an interesting challenge that would showcase and stretch their expertise, and to weed out suppliers that would be irrelevant or not committed to the process. He visited potential suppliers and gave brief presentations of the project's scope. He did not know exactly what he was looking for and focused on keeping an open mind. When suppliers were asked about their interest in participating having seen Teglgaard's presentation of the LEGO Groups vision, most were surprised by the freedom to develop their own ideas and align them with the challenges described.

The aim of this phase was to identify ten committed and capable suppliers. Suppliers were unaware of this aim; they were being asked to go beyond normal routines and processes, and some potential suppliers dropped out. However, finally ten possible suppliers were identified. Several of those that decided to participate in the project described it as an intensive experience of learning from an industrial lead user. Thus in addition to the prospects of a very large order, suppliers were motivated by this learning opportunity.

One of the reasons why they [the suppliers] enter into collaboration with us is that we often have insights into other technologies that we can offer and allow them to use this in industries that are unrelated to ours.

It is important that they [the suppliers] see some benefits from participating, if it is a one-way street, it will decrease their creativity substantially.

5.1.3. Development and delivery phases

In the beginning of the development and delivery phase, each of the ten suppliers made separate visits to the LEGO Group in Billund to present their initial ideas for a solution, and to exchange ideas with and ask questions of the relevant LEGO specialists. Two weeks before their visits, they were sent the full project brief giving them an opportunity to prepare questions and formulate initial ideas.

Based on the presentations, the LEGO packaging team selected five suppliers considered as the best prospects for becoming turnkey suppliers. Their choice was based on what they perceived to be the best understanding of the LEGO Group's ideas and vision, and needs, and the plans that seemed the most realistic for their realization. This included the ability to deliver to and service the LEGO Group's production facilities in Denmark, the Czech Republic, Hungary, and Mexico. They graded suppliers according to whether it was anticipated that they would be system suppliers or providers of critical modules for the final solution. The chosen suppliers were given three months to develop and prepare their ideas, allowing for several iterations with participation of the LEGO team. During this three-month period the LEGO team made visits to the potential supplier companies. The suppliers also arranged visits to a relevant existing customer's premises. At the end of the three months, the ideas would be presented to the final pack team in Billund, which would act as the “casting jury”. In addition to these planned meetings, Teglgaard and other members of the team were in ongoing communication with the responsible developers in the supplier organizations. LEGO's final pack team provided the development teams with additional information in the form of supplements and briefs. The competing suppliers received pallets with some 25 LEGO boxes of difference sizes and complexity with the brief that one of these boxes should be used as a case example in their final presentations. Dialogue with suppliers thus evolved around packaging issues related to this objective, in which the LEGO Group supplied the information that suppliers needed, at appropriate stages.

> *We were quite aware on how and when to provide the right knowledge for the suppliers during this phase. We did this in order to make them achieve the best way possible. We thought about this as a learning curve, where information was dripping to the suppliers and new and tightening challenges were given to them as they had absorbed the knowledge already provided.*

The five suppliers presented their ideas to the group and the steering committee over a two-week period. Each had a full day to present its ideas, discuss them with the casting panel, and receive feedback.

5.2. Next Kids Tech

Next Kids Tech is a co-creation event involving technical and engineering professionals from firms and academic organizations who were invited to participate in a one-day workshop at the LEGO headquarters in Billund together with selected LEGO employees. Next Kids Tech is one of several initiatives in the early ideation stage of the LEGO Groups innovation process and is aimed at opening up the ideation process to external input at an early stage. The workshop was organized by Technology Product Manager Gaute Munch, who heads a team of four people within the Product Technology department, responsible for the use of technology in all products across the LEGO Group's portfolio.

The first Next Kids Tech workshop took place in 2010 and was seen as successful for generating relevant and interesting contacts and inputs to the LEGO ideation process. A second workshop was held in 2011 and a third workshop is planned for 2012.

5.2.1. Before the workshop

Participants were identified from among the different LEGO employee networks, which include AFOLs (Adult Fans of LEGO), researchers, and innovation specialists. These contacts were told to extend an invitation to the workshop to whomever they thought it would be of interest. Thus the list of invitees was non-exclusive and broad, and included many people not previously known to the LEGO Group. The only requirement for participation was that the potential participant should have a specific idea or proposal for how its technology might be used in combination with existing LEGO products or technology or might be of some other use to the LEGO Group.

The invitation was quite concise, indicating the overall theme for the workshop and leaving it to potential participants to decide if and how they might contribute to the workshop. Contributions were to be tangible in the sense that participants were asked to bring prototypes, which would indicate a degree of preparation for and commitment to the task.

> *They need to bring a demo. Showing up with an idea on a piece of paper is not sufficient. They must be able to demonstrate their idea physically. Therefore they need to develop prototypes and having thought about how they want to communicate their idea to us.*

The invitation made it clear that participants could not use the workshop to get ideas, or claim ownership to ideas generated during the workshop discussion. No payment was offered for participation or in reward for any preparatory efforts or input. The invitation was for an opportunity to work with LEGO employees and others to develop inputs that might be useful for the LEGO Group. The motivation for participation was the opportunity to demonstrate a technology to decision makers in the LEGO Group, and a chance to learn from fellow participants. It might fulfill a personal ambition to contribute to further development of the LEGO brand.

> *It should be self-interest and self-realization that drives them to develop their idea and share it with us. Either enthusiasm for the LEGO brand and a desire to make it a better brand or because they have a research interest.*

The LEGO Group has received approximately 60 expressions of interest from inventors, researchers, and entrepreneurs to attend the 2012 workshop. From these, 25 to 30 potential ideas/participants will be selected. It is important that ideas are aligned to the workshop theme and the values of the LEGO Group. It is equally important to include a diversity of technologies and ideas, and for successive workshops to include new people and organizations.

> *There needs to be some wildcards. Someone, who provokes us (in the LEGO Group) and changes our way of thinking. If I selected participants for the event only based on my perception of relevance, I would limit the creative diversity.*

Prior to the workshops, the Product Technology department can support participants to improve their ideas, for example, through the provision of technical assistance or supply of LEGO bricks.

5.2.2. The workshop

A number of LEGO employees and managers from different departments are participating in the New Kids Tech project. A fundamental underlying principle of these workshops is that events take place within a tightly structured framework in order for creativity to emerge within a short time span and among people who do not know each other. Structure is provided via two means. First, the schedule is tight with relatively narrow time slots for the various assignments that participants are required to work on. The process is designed so that participants work under pressure. Second, there is an element of competition in the process, with participants required to pitch their ideas to an executive panel of LEGO senior vice presidents, to conclude the workshop. The idea of a competition is included to spur teamwork and promote enthusiasm during the workshop, not to single out a winning idea or a preferred potential partner. Rather, the purpose is precisely to generate many new contacts and maintain openness to ideas and interaction across different constellations of internal and external participants. Nevertheless, the pressure generated increases participants motivation.

> *We had invited in families and children and they were asked to make a so-called challenge session with the children. They know that at 14:30 there would be some kids with a soft drink in one hand and a cookie in the other. It puts things a bit on the edge — that you will be facing a critical and honest customer, who expresses his or her opinion directly…the evaluation panel in the evening was comprised of leading managers from the LEGO organization.*

The event kicks off with presentations by two external speakers who address different technology- or market-related themes. The participants are then organized into four groups, each of which is assigned a specific theme. The groups are organized to include LEGO employees and external participants. The external participants can choose which group to join after the workshop participants have had a chance to meet and listen to the opening presentations. This allows external participants to work on areas in which they have expertise and the chance to make the best contribution, and the opportunity for their initial inputs to shape the group's thinking.

> *Then they start to split up into these groups based on where they see ideas starting to develop in accordance with some of the things that they have brought along.*

Some people are more intimidated by working with people that they do not know and the organizers were keen that participants should feel included and as comfortable as possible.

Before starting on the group work, the participants visit the so called *demo-land*, which is a sort of exhibition where the external participants have assigned stands that allow them to present their ideas. The groups can "shop around" for inputs to the co-creation work that they will subsequently engage in. The group work takes up most of the afternoon. It is governed by templates that help to organize their work by highlighting key aspects that must be addressed irrespective of the background and perspectives of team members. "Modularizing" knowledge inputs encourages a perspective that is shared within and across groups and makes deliverables immediately comparable. The templates allow the LEGO Group to direct the work of the groups and the knowledge inputs from external team members, in directions that are most relevant for the LEGO Group. The day concludes with the LEGO committee evaluating the various ideas generated by the groups through the allocation of LEGO bricks according to criteria such as viability and feasibility of the ideas.

5.2.3. After the workshop

The workshop results in a "concept catalogue" of ideas developed by the four groups, and a detailed knowledge of the competencies possessed by each of the external participants. Some of the inputs are displayed on a LEGO Group internal Platform Opportunity Fair, where ideas from the early ideation process are presented and evaluated. The ideas developed in the workshops and the external contacts established may be taken up and developed further by different departments in the LEGO Group responsible for development projects in different areas or for different markets. The external contacts established with potential new business partners can be as important as the concrete ideas and inputs. After a day of intense working together, the LEGO Group is able to get a good grasp of participants' competencies, and the potential new partners have had an opportunity to meet and interact with people from different departments in LEGO, which allows wider assessment of their potential as future business partners.

5.3. Case comparison

The two cases in LEGO are reasonably representative of the two scenarios outlined. The packaging project is an example of a focal producer firm searching for deep knowledge and path breaking solutions within a clearly defined area. Next Kids Tech is a case of a focal producer firm seeking broader and more diverse knowledge inputs into the very early stages of its innovation process. Our case analyses reveal that the objective of an open innovation endeavor in relation to the nature of the knowledge that the producer firm seeks from external actors, implies important differences in the groups of boundary spanning activities, that is (1) selecting and mobilizing creative talent, (2) creating a shared identity and common perspective, and (3) combining and integrating knowledge. Our findings for the two projects are presented in Table 1 and discussed in the succeeding sections.

5.3.1. Selecting and mobilizing creative talent

In the individual case descriptions, various selection strategies for selecting participants are outlined. In the case of the packing project, the strategy chosen was to identify individuals and approach them at fairs or visit them at their production sites. The potential suppliers had to make the decision to commit to the idea, based on an understanding of the investments in time and resources that would be required for participation in the project. In the Next Kids Tech case, the LEGO Group published invitations in specific social media, targeting creative talent, to an informal day of creative idea exchange. This was followed up by approaches to particular individuals.

We observed that in both cases the external actors' perception of fairness and trustworthiness of the producer firm (LEGO) played an important role in mobilizing creative talent. Any co-creative endeavor carries the risk that external actors will be reluctant to share their knowledge for fear of revealing valuable intellectual property, and that the distributional and procedural fairness of the exchange relationships is not balanced. It is necessary for the focal producer firm to signal its trustworthiness as a partner. In the packaging project, which required considerable upfront investment from the involved external actors, the LEGO Group needed to signal its capability and willingness to offer a substantial return on those investments in the form of learning effects due to insights into new technological knowledge, and a possible commercially attractive, long-term business contract. In the Next Kids Tech case where significantly less upfront investment was involved and where the potential gains for the involved external actors were significantly lower, the LEGO Group had to signal via its reputation that it could provide an attractive platform that would be rewarding to the participants in the form of network contacts, intellectual inspiration, and/or enthusiasm for the brand. The focal producer firm in this case had to play the role of an effective "party host." The potential attractiveness of the "party" (i.e. the collaborative event) to each "party guest" (i.e. potentially involved external actors) is a function of the perceived attractiveness of the attracted "party crowd" (the workshop participants). Only a firm with an extremely good reputation in the market could provide a credible promise to host a party to which guests would be attracted and which would encourage networking. Of course, the LEGO brand played a major role here. In the packaging project the role was not one of "party host" but rather that of a "casting jury" which evaluated the solutions submitted based on pre-defined criteria.

In order to attract the "right" kind of external actors and to align the creative input with the corporate strategy and objectives, the formulation of the objective or problem to be solved needed to be attractive and interesting to secure the successful involvement of external actors. In the packaging project, the problem was formulated by the LEGO Group with extremely ambitious and well specified outcome parameters. This kind of problem statement served several purposes. First, it was aimed at attracting only those external actors that considered themselves able to solve the problem successfully (self-selection effect). Second, the problem statement required the external actors to stretch their resources which signaled their commitment to finding a solution that would be competitive (commitment effect). Third, the problem statement ensured that there were in place objective criteria to compare the solutions submitted which in turn increased the external actors' perception of procedural fairness (fairness effect). Fourth, this very specific problem formulation was aimed at reducing uncertainty on the side of the external actors (uncertainty reduction effect). While there were still significant uncertainties with respect to technical feasibility, their capability to solve the problem, and their chances of winning the "award," the problem statement reduced uncertainty with respect to the objectives of the solution sought.

In the Next Kids Tech case, problem formulation was kept rather open and unspecific. Apart from an indication of the theme of the workshop, potential participants were free to decide on the kind of input they wanted to contribute to the project. In order to ensure that the external actors had the relevant technological knowledge to support their ideas, they were required to submit an artifact, not just an idea. In Next Kids Tech the purpose was to attract capable individuals with relevant technological knowledge, and the problem statement was designed accordingly.

5.3.2. Creating a shared identity and common perspective

There are major differences between the two cases with respect to the group of boundary spanning activities related to shared identity and common perspective. In the packaging project, communicating performance expectations over contributions of creative solutions for a highly complex and ill-defined problem required a distinct approach. The creation of a shared identity and common perspective was an evolving process based on meetings between the LEGO Group and the candidate supplier. To convey its expectations, the LEGO Group made on-site visits to the supplier firms to explain the challenge and its corresponding contextual interdependencies and constraints within LEGO's business ecosystem. This high investment from the LEGO Group in face-to-face meetings to discuss the challenge was necessitated by the tacit nature of the problem to be solved, and its inherent ambiguity, novelty, and complexity. The LEGO Group needed the prospective suppliers to internalize the Group's perspective in order to increase the likelihood that the supplier would be able to develop a solution that would satisfy the LEGO Group's requirements. The approach involved changes to the supplier's onsite routines. It is usual for the LEGO Group to interact with suppliers' sales staff people when a new project is launched. In the packaging project, the candidate suppliers' technical staff were involved in the very early stage of introducing the project. The depth of the solution required deeper involvement of the candidate supplier firm. In the packaging project, the LEGO Group's branding played a fairly marginal role in the process of creating a shared identity.

In the Next Kids Tech case, our analysis identifies two phases related to the creation of a shared identity and common perspective: the pre-workshop and workshop phases. In the first, branding played an important role. The LEGO brand reflects the firm's core values and workshop participants – as users of LEGO products – have developed an emotional attachment to it. Thus its strong branding contributed to a shared mental model among participants with respect to the LEGO Group and its values and corresponding business objectives. The LEGO brand ensured that participants had an internalized view of the Group prior to the collaborative workshop. In the workshop setting, a well designed facilitation process was crucial for establishing focus and alignment with LEGO's strategy and objectives. This facilitation process included continuous communication of workshop objectives and deliberate interventions such as interaction with potential consumers. It was designed to achieve a balance between the aim of the LEGO Group to align the creative efforts of participants with LEGO's strategy, and the need for participants to enjoy creative freedom and independence. It became apparent that too tight enforcement of the competitive aspect on the part of the LEGO Group would crowd out the intrinsic motivation of participants and change perceptions of participants from guests at an inspiring “party,” to exploitation as cheap labor. The LEGO Group deliberately introduced elements of (inter-team) competition into the workshop design in order to increase the excitement and enjoyment of participants while increasing focus and efficiency and the alignment of the teams with LEGO's strategy and goals.

Aligning participants' expectations with those of the LEGO Group was a particular challenge in the packaging project. Here, boundary objects played an important role with the entire solution framing revolving around the “LEGO kits” issued to participants (the short descriptions of performance targets related to the packaging system, and a LEGO cardboard box), which were used as devices to communicate the problems and the issues of the LEGO Group. These were then transferred to the suppliers' project development team. In the Next Kids Tech case, the boundary objects were less well defined, which reflects the open-ended nature of this creative event. The invitation to participate in the event can be considered a boundary object. However, interviewees told us that the invitation was invariably accompanied by a series of informal discussions and conversations with the person organizing the event, and several exchanges had taken place to achieve some alignment between performance and expectations — particularly in relation to the science fair demonstration.

5.3.3. Combining and integrating knowledge

Tasks related to combing and integrating knowledge differed across cases, reflecting inherent differences in the types of ideas and underlying knowledge being sought. There were differences in the way that idea generation and combination processes were designed and governed. In the Next Kids Tech case, the primary purpose of the project is the generation of a broad set of ideas in the very early stages of new product development. The mode of collaboration chosen by the LEGO Group was one of simultaneous and multilateral user-to-user and user-to-producer interaction, a pattern seen as providing fertile ground for the generation of ideas through the provision of a platform to enable the combination of diverse sets of knowledge from different participants. In the Next Kids Tech project, idea generation is achieved primarily by teaming up creative people from within and outside the LEGO Group and providing (light) performance pressure in order to spark creative interaction. Use of templates and internal resources such as lay-outers that guide participants through the process of formulating a business proposal, means that ideas are comparable. There are two knowledge integration issues that are simultaneously present: the first concerns balancing the need for creative breadth while also ensuring that the ideas connect to the values and visions of the LEGO brand. In the Next Kids Tech case, this is a highly structured process, that uses carefully selected LEGO employees who are instructed about how to engage with external participants, to embrace their ideas, and to link them constructively to their own knowledge and ideas. At the same time, other LEGO internals work to facilitate and move the process forward, following a standardized “template” that ensures comparability of outputs in alignment with the overall plan for the day. The second integration problem in the Next Kids Tech case concerns the exchange of ideas among participants in the co-creation event. This knowledge integration task plays an indirect role but remains important to the Group insofar as it affects participants' motivation during the event and their contribution to the LEGO Group's reputation after the event. The knowledge integration issue is partially absorbed by the self-selection of external group members. External participants can choose which group to join after meeting the other participants and hearing about their contributions. Thus external participants can choose where to make their contributions and also can shape ideas and group construction before they join a group.

In the packaging project, the collaboration mode chosen by the LEGO Group is competitive bidding. Consequently, knowledge combination and integration is initially limited to the dyadic relationship between the particular candidate supplier and the LEGO Group. Once the LEGO Group has been exposed to the solutions being suggested by all the candidate suppliers, it can take on the role of knowledge architect by combining and integrating elements of different supplier-based solutions. In this project, a great deal of effort is invested in ensuring the compatibility of external suppliers with the LEGO Group's perspective. However there is a greater emphasis on the LEGO Group's ability to absorb and reflect on the inputs received from the supplier “specialists”. When preparing for the event, the LEGO team worked diligently to ensure that the LEGO organization was broadly represented in the packing team, to allow a broad range of questions to be answered or information to be retrieved quite quickly. The importance of knowledge integration became obvious to the LEGO packaging team in the course of their visits to and work with selected suppliers. The ideas for solutions proposed by suppliers pointed to internal inconsistencies and differences in viewpoints across the LEGO organization.

6. Discussion

How does this study contribute to the current literature? In general terms, while much of the extant literature on crowdsourcing for creative inputs deals with success factors at the problem or individual problem-solver level (e.g. Jeppesen & Lakhani, 2010), the governance and design of crowdsourcing activities (e.g. Boudreau & Lakhani, 2009; Pisano & Verganti, 2008; Terwiesch & Xu, 2008) or a contingency perspective of crowdsourcing (Afuah & Tucci, 2012), our study provides a complementary view by exploring organizational issues in the solution seeker's organization from a boundary spanning perspective. In this respect, our study provides a conceptual framework of crucial boundary spanning activities and applies this framework to two different case settings regarding the nature of the external inputs sought by the information seeking organization. Therefore, our study is a further attempt to contribute insights to the contingency perspective of crowdsourcing by studying how boundary spanning activities may be contingent on the nature of external input sought.

Boudreau and Lakhani (2009) suggest that there are two opposing generic design principles governing external inputs to the innovation process with "dramatically different" dynamics, with respect to participant behavior and appropriation possibilities. A community mode characterized by plus-sum games and extended collaboration among participants versus a competitive mode, characterized by prizes, rivalry, and clear-cut rules. In both LEGO cases, moments of these forms and their ensuing dynamics are identifiable. However, it is interesting that the collaborative and competitive elements are much more intertwined than implied by the notion of dramatic difference. In the Next Kids Tech case, rules and rivalry were the main mode of interaction used to add an element of enjoyment and "play" into the process, as an underlying premise for participation. This suggests that collaboration and competition is less hard-wired to the underlying arenas. Similarly, Terwiesch and Xu (2008) proposed that clever innovation contest designs – and especially the breadth of the selection pool combined with a performance-contingent award system – can control the level of inputs invested in activities where the chances of winning the contest are small. There are similarities with the breadth versus the depth of the activity and the effort invested. However, the straightforward relationship implied by Terwiesch and Xu (2008) between the reward and the rational calculation of the effort invested by participants is less visible in the cases investigated in the present paper. The roster of potential motivations seems to be broader than altruism or external (monetary) motivators, as suggested also by behavioral economics (Ariely, Bracha, & Meier, 2009). Social recognition among a group of peers seemed to be a strong motive for participation in the Next Kids Tech case, and one that cannot be framed easily as either internal or external, but seems to hinge on social acceptability rather than individual gain. A related, ongoing debate regarding innovation governance to which our case studies may contribute, relates to concerns regarding the choice between open and closed architectures of collaboration, and between hierarchical and flat governance modes (Pisano & Verganti, 2008). According to Pisano and Verganti, these dimensions constitute four modes of collaboration, each suited to different search processes and different maturity of knowledge in the host firm. The supplier case aligns well with the characteristics of the closed hierarchical form (the elite circle), in which the problems are intricate, the solution landscape is "rugged" (Terwiesch & Xu, 2008), and it is depth of experimentation and effort that is required. The Next Kids Tech project resembles that of an innovation community and a broad and non-detailed problem to which an open-ended solution is sought. However, in both cases the models evolve. In the supplier case, the definition of the problem gravitates toward the suppliers, and in the Next Kids Tech case the process could be described as transparent rather than open, with the first part of the process (inviting contributions is open), while participation in the Next Kids Tech event is based on invitation — given the participants' interest in working with a company of high standing such as the LEGO Group. This suggests that in an innovation contest, competitive and collaborative architectures can be combined in sequences, and can be regarded as phases in rather than states of collaboration.

Another important insight from our study is that the "contextuality" of the problem that is broadcasted to external actors matters. In the packaging project, "contextuality" was high since the suppliers (potential problem solvers) needed both technical knowledge and rich contextual knowledge about the LEGO production and distribution system in order to find a solution to the packaging problem. A consequence of this contextuality was that LEGO employees needed to engage in repeated and costly face-to-face interactions with employees from the supplier firms to communicate the problem in a comprehensive and appropriate way. The number of problem solvers that could be involved, therefore, was limited. Problems related to high levels of contextuality are different from problems that emerge in a crowdsourcing setting where the contextuality of the problems is low.

With respect to the organizational design issue, opening up an existing knowledge base to new inputs is problematic. Hienerth, Keinz, and Lettl (2011) point out that established organizations that mould their new product development processes around internal competences and resources need to re-organize these activities significantly in order to become "user-centric" and make room for external inputs. In the LEGO cases, this issue was dealt with directly, which confirms its importance. In both cases, the LEGO Group's management invested huge efforts to mix internal and external participants and establish "an in-between space" where external and internal providers of inputs could meet on equal terms and exchange ideas. This was most pronounced perhaps in the Next Kids Tech case, where team-building and the competition introduced facilitated the process of internal-external knowledge integration and avoided "not-invented-here" problems.

7. Implications

There are important lessons for managers involved in creating and delivering creative solutions with external partners, in how the LEGO Group managed these processes.

One is that the design of boundary spanning activities needs to be aligned to the nature of the creative input sought from external actors. Another lesson is that companies need to develop a template for the entire process, that addresses issues such as problem formulation, interaction among participants, motivation of both external and internal actors, and the design of incentives. The sophistication embodied in the design and execution of these processes in the LEGO Group points to some generic challenges for boundary spanning managers. They inspired our three generic task model which, in our view, can be applied to other industries and contexts, and which offers advice to managers about the important elements in the structuring of such processes. At a more abstract level, the mediating role of a boundary spanning manager is important in relation to creative work but also extends into a number of other areas in the network based business economy to which the generic tasks also contribute.

For academics, the contribution of this study is that it is a first attempt to explore the issue of managing cross-organizational boundary spanning processes in relation to creative work and points to some avenues for further research. For example, studies could examine the design of crowdsourcing platforms with respect to the design of incentives (intrinsic and/or extrinsic), the forms of interaction of participants (e.g. interaction versus no interaction), and problem formulations (e.g. narrow versus broad), and relate these design variables to specific outcomes (e.g. quality of ideas, number of ideas). Future research could investigate the co-existence of collaborative and competitive elements in crowdsourcing settings, and how they might best be combined (simultaneous versus sequential) to optimize intended outcomes. Further research could study how the "contextuality" of a problem impacts on the choice and effect of governance modes. Finally, it would be interesting to consider the different types of external inputs that the search organization is seeking (e.g. design ideas, physical versus virtual prototypes), and/or explicitly a performance dimension at project- or platform-level.

The LEGO cases can be considered important narratives for understanding the processes and underlying dynamics related to identifying, focusing, and integrating knowledge inputs in creative tasks. We do not claim to have discovered a blueprint for these processes that can be replicated or identified in exactly the same way in other cases. However, we believe that this paper provides a starting point for an interesting and valuable discussion of the seeming similarities in these processes. Our findings can be built on by future research.

References

Afuah, A., & Tucci, C. L. (2012). Crowdsourcing as a solution to distant search. *Academy of Management Review*, *37*(3), 355–375.

Aldrich, H., & Herker, D. (1977). Boundary spanning roles and organization structure. *Academy of Management Review*, *2*(2), 217–230.

Ancona, D. G. (1990). Outward bound: Strategies for team survival in an organization. *Academy of Management Journal*, *33*(2), 334–366.

Ancona, D. G., & Caldwell, D. F. (1992). Bridging the boundary: External activity and performance in organizational teams. *Administrative Science Quarterly*, *37*(4), 634–665.

Ariely, D., Bracha, A., & Meier, S. (2009). Doing good or doing well? Image motivation and monetary incentives in behaving prosocially. *The American Economic Review*, *99*(1), 544–555.

Augsdorfer, P. (2008). Managing the unmanageable. *Research Technology Management*, *51*(4), 41–47.

Bechky, B. A. (2003). Sharing meaning across occupational communities: The transformation of understanding on a production floor. *Organization Science*, *14*(3), 312–330.

Boudreau, K. J., & Lakhani, K. L. (2009). How to manage outside innovation. *MIT Sloan Management Review*, *50*(4), 69–76.

Burgelman, R. (1983). Corporate entrepreneurship and strategic management: Insights from a process study. *Management Science*, *29*(12), 1349–1364.

Carlile, P. R. (2002). A pragmatic view of knowledge and boundaries: Boundary objects in new product development. *Organization Science*, *13*(4), 442–455.

Choi, J. N. (2002). External activities and team effectiveness. Review and theoretical development. *Small Group Research*, *33*(2), 181–208.

Dougherty, D., & Takacs, C. H. (2004). Team play: Heedful interrelating as the boundary for innovation. *Long Range Planning*, *37*(6), 569–590.

Finch, J., & Geiger, S. (2010). Positioning and relating: Market boundaries and the slippery identity of the marketing object. *Marketing Theory*, *10*(3), 237–251.

Fleming, L. (2001). Recombinant uncertainty in technological search. *Management Science*, *47*(1), 117–132.

Florida, R. (2002). *The rise of the creative class.* New York: Basic Books.

Franke, Nikolaus, Keinz, Peter, & Klausberger, Katharina (forthcoming). Does this sound like a fair deal? Antecedents and consequences of fairness expectations in the individual's decision to participate in firm innovation. Organization Science.

Geiger, S., & Finch, J. (2009). Industrial sales people as market actors. *Industrial Marketing Management*, *38*(6), 608–617.

Goolsby, J. R. (1992). A theory of role stress in boundary spanning positions of marketing organizations. *Journal of the Academy of Marketing Science*, *20*(2), 155–164.

Grant, R. M. (1996). Toward a knowledge-based theory of the firm. *Strategic Management Journal*, *17*, 109–122.

Hall, S. (1997). The work of representation. In S. Hall (Ed.), *Cultural representation and signifying practices*. London: Sage.

Hansen, M. T. (1999). The search-transfer problem: The role of weak ties in sharing knowledge across organization subunits. Article. *Administrative Science Quarterly*, *44*(1), 82–111.

Helm, S., & Klode, C. (2011). Challenges in measuring corporate reputation. In S. Helm, C. Klode, K. Liehr-Gobbers, & C. Storck (Eds.), *Reputation management* (pp. 99–110). Berlin Heidelberg: Springer.

Hienerth, C., Keinz, P., & Lettl, C. (2011). Exploring the nature and implementation process of IT-based user-centric business models. *Long Range Planning*, *44*(5–6), 344–374.

Janowicz, S., Panjaitan, M., & Krishnan, R. (2009). Measures for dealing with competence and integrity violations of interorganizational trust at the corporate and operating levels of organizational hierarchy. *Journal of Management Studies*, *46*(2), 245–268.

Jarvenpaa, S. L., & Leidner, D. E. (1999). Communication and trust in global virtual teams. *Organization Science*, *10*(6), 791–815.

Jeppesen, L. B., & Lakhani, K. L. (2010). Marginality and problem-solving effectiveness in broadcast search. *Organization Science*, *21*, 1016–1033.

Joshi, A., Pandey, N., & Han, G. (2009). Bracketing team boundary spanning: An examination of task-based, team-level, and contextual antecedents. *Journal of Organizational Behavior*, *30*, 731–759.

Kanter, R. M. (2000). When a thousand flowers bloom: Structural, collective, and social conditions for innovation in organization. In R. Swedberg (Ed.), *Entrepreneurship: the social science view* (pp. 167–210). Oxford: Oxford University Press.

Katz, R., & Allen, T. (1982). Investigating the not invented here (NIH) syndrome: A look at the performance, tenure, and communication patterns of 50 R&D project groups. *R&D Management*, *12*(1), 7–19.

Katz, D., & Kahn, R. L. (1978). *The social psychology of organizations.* New York: John Wiley & Sons.

Keller, R. T., & Holland, W. E. (1975). Boundary-spanning roles in a research and development organization: An empirical investigation. *Academy of Management Journal*, *18*(2), 388–393.

Kellogg, K. C., Orlikowski, W. J., & Yates, J. (2006). Life in the trading zone: Structuring coordination across boundaries in postbureaucratic organizations. *Organization Science*, *17*(1), 22–44.

Kragh, H., & Andersen, P. H. (2009). Picture this: Managed change and resistance in business network settings. *Industrial Marketing Management*, *38*(6), 641–653.

Kristensen, P. S. (1992). Flying prototypes: Product departments' direct interaction with customers. *International Journal of Operations and Production Management*, *12*(7/8), 197–212.

Levina, N., & Vaast, E. (2005). The emergence of boundary spanning competence in practice: Implications for implementation and use of information systems. *MIS Quarterly*, *29*(2), 335–363.

Lichtenthaler, U., & Ernst, H. (2006). Attitudes to externally organising knowledge management tasks: A review, reconsideration and extension of the NIH syndrome. *R&D Management*, *36*(4), 367–386.

Lysonski, S. J., & Johnson, E. M. (1983). The sales manager as a boundary spanner: A role theory analysis. *Journal of Personal Selling & Sales Management*, 8–21.

Mantere, S. (2008). Role expectations and middle manager strategic agency. *Journal of Management Studies*, *45*(2), 294–316.

Marrone, J. A., Tesluk, P. E., & Carson, J. B. (2007). A multilevel investigation of antecedents and consequences of team member boundary spanning behaviour. *Academy of Management Journal*, *50*(6), 1423–1439.

Martins, E. C., & Terblanche, F. (2003). Building organisational culture that stimulates creativity and innovation. *European Journal of Innovation Management*, *6*(1), 64–74.

Miles, M. B., & Huberman, A. M. (1994). *Qualitative data analysis — A sourcebook of new methods* (2nd ed.). : Sage Publications.

Orlikowski, W. J. (2002). Knowing in practice: Enacting a collective capability in distributed organizing. *Organization Science*, *13*(3), 249–273.

Perry-Smith, J. (2006). Social yet creative: The role of social relationships in facilitating individual creativity. *Academy of Management Journal*, *49*(1), 85–101.

Pisano, G. P., & Verganti, R. (2008). Which kind of collaboration is right for you? *Harvard Business Review*, *86*(12), 79–86.

Ratcheva, V. (2009). Integrating diverse knowledge through boundary spanning processes — The case of multidisciplinary project teams. *International Journal of Project Management*, *27*, 206–215.

Rosenkopf, L., & Nerkar, A. (2001). Beyond local search: Boundary-spanning, exploration, and impact in the optical disk industry. *Strategic Management Journal*, *22*, 287–306.

Schultz, M., & Hatch, M. (2003). The cycles of corporate branding. *California Management Review*, *46*(1), 6–26.

Shalley, C. E., & Gilson, L. L. (2004). What leaders need to know: A review of social and contextual factors that can foster or hinder creativity. *The Leadership Quarterly*, *15*, 33–53.

Singh, J. (1993). Boundary role ambiguity: Facets, determinants, and impacts. *Journal of Marketing*, *57*(2), 11–32.

Stake, R. E. (1995). *The art of case study research.* Thousand Oaks, CA: Sage Publications.

Strauss, A., & Corbin, J. (1990). *Basics of qualitative research: Grounded theory procedures and techniques.* : Sage Publications.

Teigland, R., & Wasko, M. M. (2003). Integrating knowledge through information trading: Examining the relationship between boundary spanning communication and individual performance. *Decision Sciences*, *34*(2), 261–286.

Terwiesch, C., & Xu, Y. (2008). Innovation contests, open innovation, and multiagent problem solving. *Management Science*, *54*(9), 1529–1543.

Tushman, M. L. (1977). Special boundary roles in the innovation process. Article. *Administrative Science Quarterly*, *22*(4), 587–605.

von Hippel, E., & von Krogh, G. (2003). Open source software and the 'private-collective' innovation model: Issues for organization science. *Organization Science*, *14*(2), 209–223.

Woodman, R., Sawyer, J., & Griffin, R. (1993). Toward a theory of organizational creativity. *The Academy of Management Review*, *18*(2), 293–321.

Zaheer, A., McEvily, B., & Perrone, V. (1998). Does trust matter? Exploring the effects of interoganizational and interpersonal trust on performance. *Organization Science*, *9*(2), 141–159.